

Andrew Marion

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EDUCATION

KENNESAW STATE UNIVERSITY (KSU) | Marietta, GA

Anticipated Graduation May 2027

• *Bachelor of Science in Mechanical Engineering, Aerospace Engineering Minor*

GPA: **3.81/4.0**

• Relevant Coursework: Aerodynamics, Aircraft Design, Fluid Dynamics, Heat Transfer, Computer Aided Engineering

RELEVANT EXPERIENCE

Textron GSE | Cartersville, GA

June 2026 – August 2026

Summer 2026 Mechanical Engineering Intern

- Supported new product development for the electrification of airport ground support equipment (GSE), including tow tractors and belt loaders, contributing to Textron's shift toward electrified vehicle platforms
- Designed assemblies in Siemens NX and created engineering drawings, including a hinged door for the MX4 electric baggage tractor and access doors for the 660-belt loader electrification refresh, delivering production-ready components
- Conducted simulations and physical prototype testing, including sensor location testing for parking brake actuation and simulating access door motion, verifying correct operation and reducing design rework before production release

RESEARCH EXPERIENCE

KSU Advanced Aerodynamics and Propulsion Lab | Marietta, GA

September 2023 – Present

Undergraduate Research Assistant

Numerical Investigation of Compound Delta Wings

September 2023 – Present

- Executed 140+ CFD simulations in Ansys Fluent to characterize aerodynamic performance across velocities and angles of attack, improving lift-to-drag efficiency from 7.24% to 13.57% in the transonic regime
- Authored and presented first peer-reviewed research paper at the 2025 AIAA Region II Student Conference; demonstrated technical writing and oral communication skills essential in engineering environments
- Applied high-fidelity turbulence modeling (Large Eddy Simulation) and GPU-accelerated computing to validate aerodynamic performance predictions, reducing simulation error and improving result reliability

Multi-Fidelity Assessment of Mid-Fidelity Control Surface Modeling with FlightStream

March 2025 – Present

- Evaluated mid-fidelity control surface deflection modeling in FlightStream to assess accuracy boundaries for use in preliminary aircraft design, in collaboration with S2FAR Lab at Oklahoma State University and Siemens
- Co-authored findings published at AIAA SciTech 2026, validating high-fidelity simulations against physical flight testing and WindShaper wind tunnel data to ensure computational models meet real-world accuracy standards

Aerodynamic Characterization of a Blended Wing Body Using FlightStream

January 2026 – Present

- Benchmarked FlightStream against RANS-based CFD models in collaboration with Siemens to assess predictive accuracy of flow behavior over blended wing body aircraft, resulting in 2 papers accepted to AIAA SciTech 2027
- Assessed FlightStream's ability to accelerate simulation turnaround for novel aircraft concepts, enabling faster design iteration in early-stage development

COMPETITION TEAMS

Aerial Robotics Competition Team | Marietta, GA

September 2023 – April 2025

PADA Team Lead

- Led 20+ members through full design, manufacturing, and flight testing lifecycle of a powered autonomous delivery aircraft (PADA) for SAE AERO Design
- Utilized ArduPilot to enable autonomous flight of a custom-designed flying wing, which detached from a larger carrier aircraft mid-flight and autonomously navigated to landing coordinates detected by the parent aircraft
- Configured a Matek H743 flight computer and tuned its autopilot to achieve stable flight on an inherently unstable platform

Kennesaw Rocketry Team | Marietta, GA

December 2025 – July 2026

Vice President

- Leading administrative team in organizing public events and standardizing documentation; previously Liquid Propulsion Lead directing injector and combustion chamber development
- Conducted high fidelity CFD of propellant injection and air brake deployment supporting rocket systems development

TECHNICAL SKILLS

CAD Software: SolidWorks, NX, AutoCAD, and ANSYS SpaceClaim

CFD Software: ANSYS: Fluent, SolidWorks Flow Simulation, OpenVSP, FlightStream, and ParaView

Programming Language/Autopilot Software: MATLAB, Python, ArduPilot, LaTeX, and R

Industry Tools: 3D Printer, Laser Cutter, Composite Layup, Lathe/Mill, CNC Machining and Foam Cutter